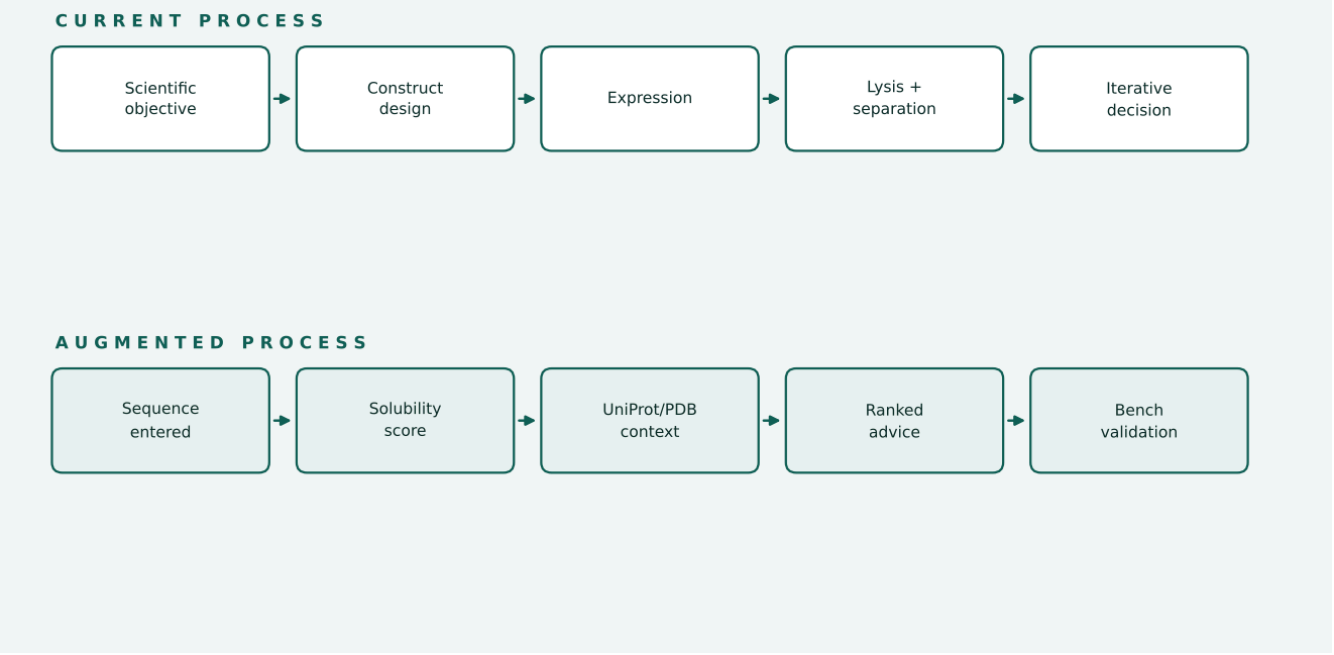
OPENCLASSROOMS / DATA & MACHINE LEARNING PROJECT MANAGEMENT

Data & ML Project Management Report

Predicting protein solubility: supporting construct selection and laboratory experiments



Fatima ADDA-REZIG

Individual project for the Data Scientist – Machine Learning programme | Field: biotech / protein production

Project status: demonstrator published; documentation and evaluation evidence updated as of 29 October 2026.

Report written according to the Data & ML project management template provided for the programme. Business-oriented analysis, steering decisions and verifiable evidence. Scientific detail in the appendices. Layout and appendix revision: 5 November 2026; scientific metrics unchanged.

EXECUTIVE SUMMARY

A better-informed decision before the bench

The problem. In protein production, a construct can look promising on paper yet yield too little soluble material to be studied. Failed experiments consume cultures, consumables and scientific time. Today, construct selection relies largely on experience and on successive rounds of testing.

The proposal. I built a tool in which the researcher pastes a protein sequence. They obtain an estimate of its propensity to be soluble, a biological identification where public databases allow a reliable match, then a reasoned list of experiments to compare: plain construct, purification tag, expression host and functional controls.

The outcome. After several versions and comparisons, the demonstrator is available online, with automated tests and documented source code. The best model tested better separates soluble from insoluble proteins on the set used for its development: its discrimination indicator is approximately **0.83 on a scale from 0.5 (chance)**

to 1 (perfect). This measurement is not sufficient to prove a real laboratory gain.

The decision. Stop developing the prototype, keep a stable version and prepare an independent biological pilot. Success in the pilot will not be prediction quality alone: it will need to show that the tool leads to better experiment choices, at an acceptable cost and turnaround.

Decision-maker's question	Current answer
What is it for?	Prioritising which constructs to test, without replacing experiments.
What result exists?	Working demonstrator; development score ≈ 0.83.
What is not yet established?	The actual reduction in experiments, time and spending.
What is the next decision?	Pilot on new constructs with traceable results.

Reading convention: measured results = taken from test outputs; estimates = economic assumptions; future project = steering proposal. Full values and methods are available in the appendices.

HOW TO READ THIS REPORT

Useful concepts without the jargon

Understanding it in under two minutes

Term encountered	Operational meaning
Predictive model	A program that learns to classify sequences from examples.
Score from 0 to 1	An indication of solubility: it is not a purification yield.
Decision threshold	The value above which the software says « soluble ».
False positive	Predicted soluble, but insoluble in the data: risk of a pointless experiment.
False negative	Predicted insoluble, but soluble in the data: a possibility discarded.
Discrimination quality (AUC)	The model's ability to rank a true soluble ahead of a true insoluble.
Calibration	Checking that displayed scores are consistent with observed frequencies.

Three levels of evidence, used throughout

- **Measured:** data, Git dates, test results and verified deliverables.
- **In development:** performance useful for comparing options, but not certified on a new cohort.
- **Assumption or target:** budget estimate, expected gain or threshold proposed for a pilot.

Scope of the project presented

The delivered product is an **experimental decision-support tool**. The software does not directly determine the best yield, the best tag or the best temperature; it organises comparisons that still have to be validated at the bench.

A fuller glossary is provided in Appendix A; the summary above deliberately avoided technical acronyms.

Contents and correspondence with the template

Assessed parts	Report pages
1. Needs and context · 2. Data solution audit	3 to 6
3. Solution choice · 4. Strategy and method	7 to 11
5. Monitoring and control · 6. Conclusion	12 to 14
7. Appendices, detailed results and sources	15 to 22

1. CONTEXT AND NEEDS ANALYSIS

1.1 Organisation and data & ML maturity

Organisation and sector

This is an **individual project** carried out for the Data Scientist – Machine Learning programme, applied to R&D in biochemistry, biotechnology and recombinant protein production. The project « organisation » consists of one person covering scoping, data preparation, modelling, development and documentation. There is neither an industrial client commissioning the work nor laboratory staff assigned to it. The biologist, platform manager and quality roles correspond to target contacts for a possible pilot, not to a formally committed team.

Starting maturity: targeted SWOT assessment

Dimension	Observed strengths	Weaknesses / watch points
Domain / data	Hands-on experience of production, purification and quality control	Heterogeneous public data; incomplete production conditions
Data & ML	First supervised pipeline, metrics and tests	Sensitivity to homology; weak external evidence
MLOps	Git, FastAPI, Docker, CI/CD, tracking and optimisation explored	Gap between technical prototypes and real scientific use
Adoption	Concrete problem, identifiable user profiles	No measurement of usage or experimental gains
Security	Reproducibility and separation of private artefacts	External sequence search to be governed

Strategic issues linked to the data

The maturity sought is not autonomous industrial production, but an **operational and controllable demonstrator**: easy input, data provenance, explicit threshold, understandable recommendations, traceable code and stated limitations. Moving to a laboratory service would require a prospective pilot and confidentiality rules.

1. CONTEXT AND NEEDS ANALYSIS

1.2 Business need, users and criteria

Three field observations that guided the scoping

- Selection is made by comparing domains, construct boundaries, mutations and tags according to the experimental objective.
- Recurring difficulties: precipitation, aggregation, misfolding, insufficient soluble fraction or loss of activity.
- An unsuccessful cycle commits preparation, expression, lysis, analysis and sometimes purification; costs vary with the target and the scale.

Stakeholders and priority requirements

Target user	Need	Success criterion
Biochemist / research engineer	Decide which experiments to launch	Readable score and reasoned proposals
Platform manager	Compare the cost of options	Cost tracking and experiments avoided
Data/ML project lead	Compare, reproduce, deploy	Versions and regression tests
Quality / IT	Control data and third-party services	Traceability, access control and risk assessment

Needs gathering: drawing on professional experience and a review of the production steps; user interviews and cross-team workshops remain future work. Prioritisation comes from individual scoping, not from a client consensus.

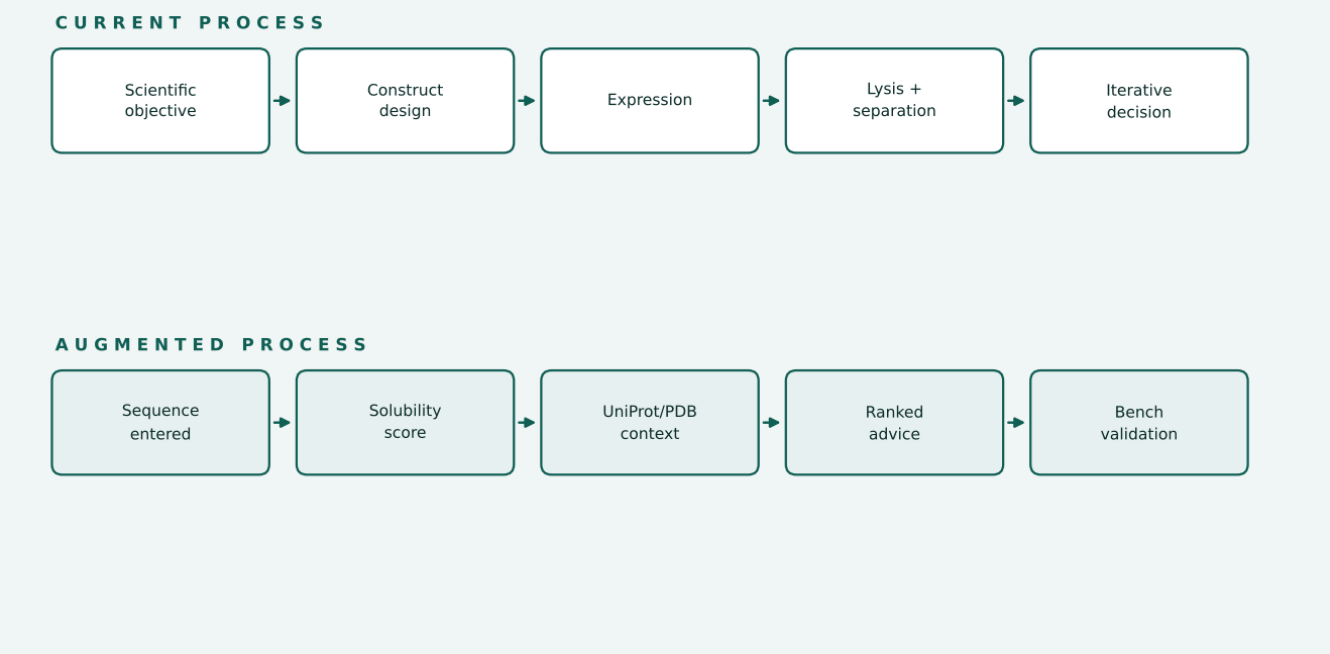
Functional requirements retained

- Essential: predict from a raw sequence, state the length analysed, freeze the parameters and return a verifiable result.
- Important: identify the protein with provenance, propose ranked constructs, produce a PDF sheet.
- Out of scope: guaranteeing an optimal tag, yield, host or temperature without dedicated data.

2. DATA SOLUTION AUDIT

2.1 Current process and data flow

The usual process is iterative: the researcher defines a target, designs several constructs, runs experiments, measures the soluble and insoluble fractions and starts again if needed. The tool adds a decision-support step **ahead of the first experiment**, without removing experimental validation.



Data flow put in place

Input / step	Transformation	Output / control
Raw sequence	Check of the 20 standard AA, length	Admissible input; explicit errors
Labelled data	Identifier alignment and homology control	Separate development partitions
Sequence model + Bio13	Encoding and ranking by the model	Score and length actually analysed
Public databases	PDB/UniProt similarity search	Annotation with provenance, where reliable
Business rules	Ranking of experiments and controls	Indicative experimental plan + PDF
Code versions	GitHub tests + private checkpoint management	Reproducibility and traceability

Risks associated with the process

A sequence may be confidential; an external search requires an authorisation policy. Public solubility data do not always describe the same expression protocol, and two homologous proteins can distort the evaluation if they move from one set to another.

2. DATA SOLUTION AUDIT

2.2 Data, partitions and discrepancies

One dataset, five distinct uses

Partition	Count	What is it for?
Fit	46,270	Learning the model parameters.
Tune	4,052	Comparing and choosing variants during development: exploratory AUC and PR-AUC, not independent.
Calibration	4,052	Internal validation of fine-tuning: choice of the optimal epoch and early stopping. Already consumed; no final probability calibration on this set.
Threshold	4,052	Split into two distinct groups of 2,026 proteins each (approximate homology): Platt calibration on the first, threshold selection on the second; no independent final test.
Holdout	4,052	Set held back; not an independent test available for the fine-tuned version.

Why separate them? Fit (46,270 proteins) is used to learn the model parameters. Calibration (4,052) was used during the campaign to select the epoch and early stopping of the fine-tuning: it is therefore already consumed. Tune (4,052) guided the choice of architectures and cannot be presented as an independent test. At finalisation, Threshold (4,052) was split into two distinct subsets of 2,026 proteins each, following approximate homology groups: one fits the Platt calibrator on the logits, the other selects the threshold under a recall constraint. Both subsets take part in the tuning; neither constitutes an independent final test. The MMseqs2 audit is approximate and does not exclude every possible homology.

Discrepancies found during the audit

Point audited	Risk to the business objective	Project response
Labels and protocols	A « soluble » annotation may depend on host and protocol	Document the scope of the score
Homologous sequences	Artificially inflated performance	Control of similarity between groups
Score / yield confusion	Overconfident decisions	State that a score does not measure yield
Long truncation	Loss of useful domains	Documented length policy
No business measurement	Unverifiable ROI	Plan a controlled pilot

3. IDENTIFYING A TARGET SOLUTION

3.1 Use cases: prioritisation method

The use cases were identified from the decision points of the process described in section 2.1, then scored on four criteria: **business impact 40%**, **feasibility 30%**, **data availability 20%** and **risk control 10%**. Each score runs from 1 (low) to 5 (high). This scoring is a scoping proposal, not the outcome of a client committee.

Use case	Impact	Feas.	Data	Risk	Score /5	Rank
Sequence ranking	**5**	**4**	**5**	**3**	**4.5**	**1**
Target identification	**4**	**4**	**4**	**3**	**3.9**	**2**
Experiment plan and controls	**4**	**3**	**2**	**2**	**3.1**	**3**
Tag/host/induction optimisation	**5**	**1**	**1**	**1**	**2.6**	**4**

Product trade-off

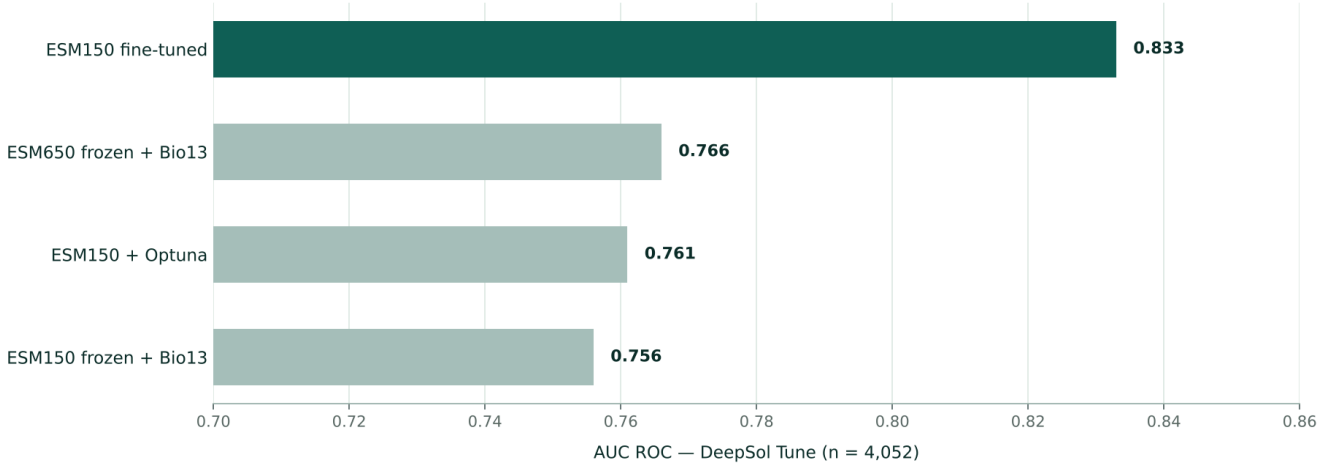
- **Priority 1:** ML ranking of sequences with readable results.
- **Priority 2:** automatic identification, with a confidence threshold and sources.
- **Priority 3:** experimental advice based on expert rules, with conditions and contraindications.
- **Deferred:** quantified optimisation of tags, temperature or host, for lack of suitable training data.

The delivered product reflects this trade-off: expression advice is not presented as a second validated prediction, but as guidance for building a comparison plan.

3. IDENTIFYING A TARGET SOLUTION

3.2 Comparison of technical options

The trade-off considered **discriminative power**, **compute cost**, **ease of deployment** and **relevance to protein sequence**. The first LightGBM solution was inexpensive and production-ready, but limited in performance. Pre-trained protein representations were tested next, then adapted to the task.



Option	Performance on Tune	Decision
ESM-2 150M frozen + 13 Bio13 descriptors	≈ 0.76	Lightweight alternative
Same model + hyper-parameter optimisation	≈ 0.76	Insufficient gain
ESM-2 650M frozen	≈ 0.77	More expensive to compute
ESM-2 150M partially re-trained	≈ 0.83	Retained for the demonstrator

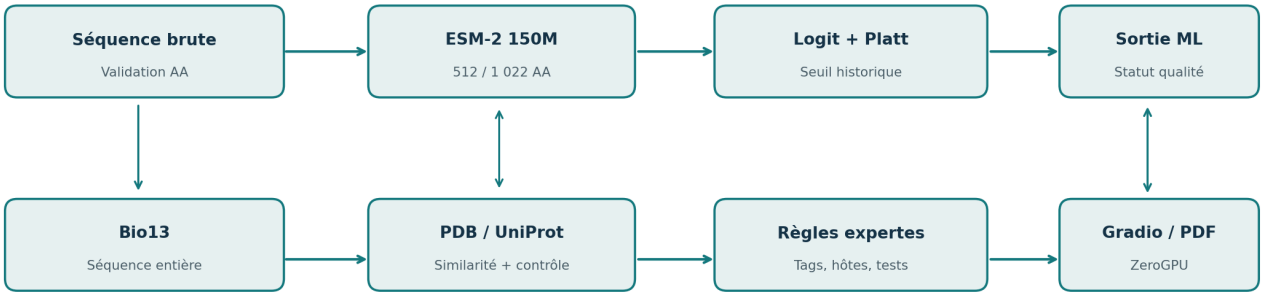
AUC ROC measures the model's ability to separate soluble from insoluble proteins. The development result AUC ROC = 0.832857 (PR-AUC / average precision = 0.786022) was obtained on Tune (n = 4,052), a set already used to compare and select variants. It is neither a final independent performance figure nor proof of experimental usefulness. A versioned retrospective export (predictions, metrics by length band, manifest and SHA-256) is now published in resultats/evaluation_v1/; the method and its limitations are detailed in appendices E and F.

What we cannot conclude

The improvement cannot be attributed to a single setting: the classification model, the fine-tuning and the pipeline all changed together. The best score describes the prototype selected; on its own it does not measure economic value in the laboratory.

3. IDENTIFYING A TARGET SOLUTION

3.3 Architecture retained and constraints



The product separates **three functions**: 1) a prediction produced by training; 2) a PDB/UniProt search providing biological context where the match is strong enough; 3) a rules engine that ranks the possible constructs and controls. This separation avoids claiming that a recommended tag is predicted by the neural network.

Length policy applied

Protein length	Information used	Display
Up to 512 AA	The whole sequence	Reference mode
From 513 to 1,022 AA	The whole sequence	Experimental mode, score not recalibrated for this mode
Beyond 1,022 AA	First 512 AA	Truncation flagged

The development test shows particular value for sequences of 513 to 1,022 AA (**360 examples: AUC ≈ 0.81 → 0.85**). The overall gain is small, because most proteins are shorter. This finding supports a cautious extension rather than a silent change to the model.

Delivery

Gradio web interface on Hugging Face Spaces / ZeroGPU, documented code on GitHub, private weights in a separate repository and automated tests without GPU. Robustness under load and the use of proprietary sequences remain out of scope.

4. STRATEGIC AND METHODOLOGICAL SUPPORT

4.1 Method, pivot and governance

Project management: iterative approach inspired by CRISP-DM

The project followed short loops: **frame the need**, audit the data, build a baseline, compare, analyse errors, then improve the delivery. Decisions are backed by experiments, commits and a change log. This method makes it possible to redirect effort once the real need becomes clearer.

The change of direction is a steering decision

Phase 1 — industrialisation first. The initial project emphasised LightGBM, FastAPI, Docker, continuous integration, tests and monitoring. It demonstrated a complete MLOps pipeline, but the product remained centred on a prediction with little context for the biologist.

Phase 2 — usage and prediction quality first. Experiments with protein models showed a gain on the development set. I therefore retained fine-tuned ESM-2, replaced the demonstration interface with Gradio and added the biological annotations and design rules.

What was kept: Git, versioning, tests, separation of artefacts, tracking logic and reproducibility requirements. **What was set aside:** the Docker/ONNX API as the main product; its code is archived, not lost. This pivot favours the quality of the business demonstration without erasing the MLOps skills acquired.

Decision-making arrangements

The project owner combines business and technical responsibilities in this individual POC. In a pilot, the biologist would validate the use cases and results, the platform manager would arbitrate the budget, the Data Scientist would run the evaluations, and the IT/quality lead would control access and confidentiality.

Roadmap: dates, milestones and responsibilities



Available history: planned and actual

Milestone	Planning reference	Actual / evidence	Variance
First MLOps pipeline	Initial objective: API and tests	03–05/07/2026: FastAPI, Docker, CI commits	Milestone met; initial duration not recorded
Initial V2/V3 cycle	Estimated effort reference: ≈ 15 person-days	03–24/07: 22 calendar days between first and last commits	Not computable: person-days ≠ calendar days
ESM-2 campaign	Original dated forecast not retained	October: documented comparisons and fine-tuning	Not computable
Demonstrator and repository stabilised	Version freeze decision	22–25/10: Space, tests, GitHub version	Delivered on 25/10

The actual dates come from the commit history and experiment outputs. **No earlier quantified contractual plan was found**: inventing a precise delay would be misleading. The table therefore compares the starting objectives with the dates actually documented.

Prospective plan proposed for business validation — not carried out

Proposed phase	Indicative window	Target owner	Exit milestone
Define cohort and protocols	Week 1	Biologist + Data	Inclusion criteria signed off
Prepare and audit the data	Weeks 2–3	Data + quality	External set frozen
Comparative laboratory screening	Weeks 4–6	Biologist / platform	Soluble/insoluble results and activity
Cost, lead-time and adoption review	Week 7	Project lead + manager	Go / no-go decision

4. STRATEGIC AND METHODOLOGICAL SUPPORT

4.2 Financial trade-offs and risks

Three funding scenarios — estimates, not commitments

Scenario	Scope	Estimated budget
A — data quality	Quality, homology, protocol and traceability	€7,000
B — ML improvement	A + new evaluations and calibration	€12,500
C — biological pilot	B + prospective trial and value measurement	€23,000

Transparent costing, cumulative assumptions excl. VAT: A = 14 person-days (€6,500) + tools/storage (€500), i.e. €7,000. B = A + 9 person-days (€4,800) + GPU/infrastructure (€700), i.e. €12,500. C = B + 7 person-days (€3,500) + 10 screening experiments covering 10 constructs (one exploratory experiment per construct, €600 of direct costs each, i.e. €6,000) + a separate contingency reserve (€1,000), i.e. €23,000. Daily rates, the scope of the experiments and the cost items are set out in Appendix E. These are steering estimates, not quotes, committed spending or observed savings.

A purely illustrative return-on-investment example: if one complete experimental cycle avoided were worth €2,000 (to be checked with a laboratory), the break-even points would be 4, 7 and 12 cycles avoided respectively. This full cycle cost is not the same as the €600 exploratory experiment in scenario C, nor the same as the model's FN/FP weightings; no actual saving has been measured.

The cost function: a debatable and explicit business trade-off

The threshold currently frozen uses **€1,200 per false negative** (a genuinely soluble protein discarded) and **€600 per false positive** (a pointless experiment on an insoluble one). It therefore favours recovering potentially interesting proteins, even at the price of extra experiments. This choice is coherent only if **the value of a missed good target exceeds the cost of an avoidable experiment**.

If the priority objective becomes strictly the reduction of pointless experiments, this weighting must be discussed, and possibly reversed, **before** any new threshold tuning on separate data. The prototype does not claim to show that €1,200 and €600 are a laboratory's true costs.

Main risks and responses

Risk	Impact	Mitigation
Data leakage through homology	Artificially optimistic score	Grouping and similarity search
Labels dependent on protocols	Poorly transferable decisions	Document the reference conditions
Proprietary sequence sent to a third party	Confidentiality	Disable external search without authorisation
Advice taken as guaranteed predictions	Unsuitable experiments	Explained rules + scientific validation
No economic monitoring	ROI not provable	Measurement planned per experimental cycle

Success indicators (business and technical KPIs)

Category	KPI to track	Evidence / status
Business	**Experiments avoided; soluble fraction; hours and € per construct**	**Pilot with a control batch; not measured**
Model	**AUC ≈ 0.83 on Tune; recall ≈ 0.86 at the chosen threshold**	**Development measurements**
Service	**Calibration, drift, P95 latency, availability**	**To be assessed in operation**

Scientific responsibility, confidentiality and compliance

The decision to launch an experiment remains human: no production choice can rest on the score alone. In a pilot, any transfer of confidential sequences to PDB/UniProt requires authorisation; licences, terms of use and external annotations must be checked. A biologist confirms the recommendations and the experimental results.

5. MONITORING AND CONTROL

5.1 Dashboard: lead times, costs, deliverables, quality

The dashboard distinguishes an **observed situation** from an **objective to be steered**. A « not recorded » cell means that no reliable historical measurement is available, not that the cost or the delay is zero.

Dimension	Planned / objective	Actual or observed status	Variance / decision
Lead time – MLOps cycle	Estimated effort: 15 person-days	3–24 July: 22 calendar days	Not comparable; time spent not recorded
Lead time – demonstrator	Version stabilised in October	application delivered, GitHub and CI as of 25/10	Deliverable met
Costs – project delivered	Initial reference budget not documented	Actual spending: not recorded	Set up invoice/time tracking
Costs – future pilot	Scenarios: €7,000 / 12,500 / 23,000	No budget committed	Submit for arbitration
Deliverables	Interface, documentation, tests	Application online, PDF, repository and Actions	Complete for the demonstrator
Data quality	Separate partitions, homology controlled	62,478 sequences partitioned	Exhaustive control not guaranteed
ML performance	Best discriminator on Tune	AUC 0.832857; PR-AUC 0.786022 on Tune (development)	No external validation of the fine-tuning at this stage
Simulated error cost	Minimise FN/FP cost with a minimum recall	€387,600 on the threshold-selection subset	A tuning measure, not an observed saving

Reading the indicators: costs, time and deliverables

Actual lead time: the time elapsed between two commits is not a measure of days worked. **Actual cost:** without timesheets, invoices and infrastructure costs, no monetary « actual » can be declared. **Performance:** a technical score does not replace measuring the experiments saved.

This table avoids mixing a prospective budget envelope, a hypothetical misclassification cost and spending actually borne by the project.

5. MONITORING AND CONTROL

5.2 Tools, reporting and delivery quality

Tools actually used

Tool	Purpose	Evidence / limitation
Git / GitHub	History, branches, documentation	Dated commits, main version stabilised
GitHub Actions	Run the regression tests	Latest tests passing without GPU
Google Colab / GPU	Train and compare the models	Campaign outputs saved
Hugging Face Spaces	Web demonstration	Gradio accessible; latency to be characterised
MMseqs2	Search for related sequences	Approximate audit, not exhaustive
PDB / UniProt	Place the score in context	Match to be verified

Monitoring cadence recommended during a pilot

- At every data import: quality control and search for duplicates and homologies.
- At every commit: automated tests and secrets check.
- Every week: milestone progress, committed spending, hours consumed and incidents.
- After every campaign: soluble fraction, activity, experiment avoided or not, full cost and user feedback.

Key success factors and watch points

Gate	Condition to pass
Scientific	Results reproduced on unseen proteins and comparable protocols
Business	Proven reduction of pointless cycles without excessive loss of good targets
Operations	Confidentiality, availability and maintenance documented

6. CONCLUSION AND RECOMMENDATIONS

A finished demonstrator, a pilot still to be decided

The project moved from an initial classification and industrialisation pipeline towards a research application that is more directly usable: the user enters a sequence, sees a prediction, obtains biological context where available and prepares a comparative experiment plan.

The version retained, ESM-2 150M partially fine-tuned with Bio13, shows the best results among the variants compared on Tune (AUC ROC 0.832857; PR-AUC 0.786022), but that set was used for development and is not an independent validation. The retrospective exports, artefact fingerprints and evaluation script make these

computations inspectable, without constituting a full re-run of the checkpoint or an external homology check. The next step proposed, not carried out, is a frozen evaluation on a new cohort of proteins, with homology control and a calibration check, particularly between 513 and 1,022 AA, before a biological pilot.

Decisions taken while running the project

- Abandon the raw addition of new external proteins when performance drops on the comparison set.
- Prefer a partially adapted protein model over a heavier frozen architecture that is more expensive to run.
- Keep the MLOps components acquired in the history, while refocusing the product on a business interface.
- Present tag/host recommendations as explainable business rules rather than as a learned optimisation.
- Freeze the delivered product in order to stop successive adjustments on the same data.

Next decision proposed

Authorise, or not, a frozen prospective evaluation before any pilot: checkpoint, calibrator, threshold and protocol frozen; a genuinely new cohort; control of homologous sequences; AUC/PR-AUC, Brier/ECE and confusion matrices reported with counts; specific analysis of the 513–1,022 AA range and recalibration if needed. This validation is a prospect; it has not yet been carried out. A subsequent biological pilot will have to track yields, activity, spending and both types of error in order to decide on a go/no-go.

7. APPENDICES

Appendix A — operational glossary

Term	Definition for a non-specialist reader
AA	Amino acid: a unit of a protein sequence.
AUC ROC	A measure between 0 and 1 of ranking quality; 0.5 is equivalent to chance.
PR-AUC	A measure of ranking quality for the soluble class, sensitive to its frequency.
Bio13	Thirteen biochemical features computed on the sequence.
Calibration / Platt	A method for relating raw scores to observed frequencies.
CI/CD	Automated tests and operations around code changes.
ESM-2 150M	A protein language model with roughly 150 million parameters.
Fine-tuning	Adapting part of a pre-trained model to a specific task.
FN / FP	FN: a soluble protein rejected; FP: an insoluble one accepted.
Gradio / ZeroGPU	Web interface / on-demand GPU access mechanism.
Homology	Kinship between proteins that may make samples dependent.
LightGBM	A classification method for tabular data.
MMseqs2	Software for fast similarity search between sequences.
PDB / UniProt	Public databases of structural and functional information on proteins.
Tag / host	A label fused to the protein / the organism or cell used to produce it.
Holdout / Tune	Set held back for testing / set used to choose between models.
MLOps	Practices for versioning, testing, shipping and monitoring ML models.

7. APPENDICES

Appendix B — quantified results and options discarded

Model comparison — on the Tune set (n = 4,052)

Variant explored	Exact AUC	Exact PR-AUC
ESM-2 35M frozen	0.7297	0.6587
ESM-2 35M + Bio13	0.7383	0.6735
ESM-2 150M frozen	0.7511	0.6854
ESM-2 150M + Bio13	0.7562	0.6864
ESM-2 150M + Optuna	0.761432	0.693242
ESM-2 650M frozen + Bio13	0.766446	0.701039
ESM-2 150M fine-tuned + Bio13	0.832857	0.786022

Additional inputs not retained

UEsolDS: 70,029 unique valid sequences, then exclusion of 10,994 exact duplicates and 25,386 identified homologues; 33,649 further sequences retained. The Tune AUC of the frozen model + Bio13 fell from 0.756177 to 0.733012 after increasing Fit to 79,919. This drop does not on its own prove its cause.

Additional descriptors: best Tune variation observed +0.0014 with raw length, too small to justify a change.

Truncation: Tune figures

Population	N	AUC 512 AA	AUC 1,022 AA
Total	4,052	0.832857	0.834727
Up to 512 AA	3,672	0.834522	0.834522
513 to 1,022 AA	360	0.813852	0.848630

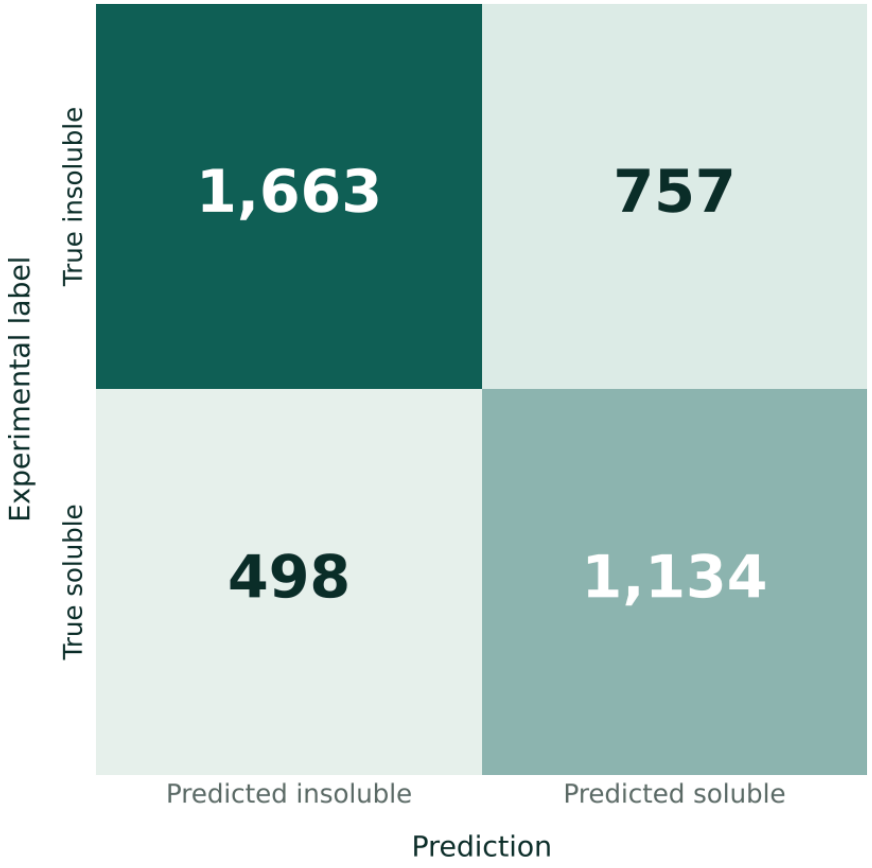
The 20 proteins longer than 1,022 AA are not shown graphically: the sample is too small to support a stable interpretation.

7. APPENDICES

Appendix C — classification errors and economic threshold

Confusion matrix available: earlier frozen ESM-2 variant

The example below is a real measurement of an earlier frozen ESM-2 150M variant with Bio13, on its historical holdout of 4,052 proteins. It is not a confusion matrix of the fine-tuned version. Comparing it directly with the delivered version would be incorrect.



Error / historical result	Count	Business meaning
True negatives	1,663	Insoluble proteins correctly set aside
False positives	757	Insoluble judged soluble → possible pointless experiments
False negatives	498	Soluble proteins set aside → missed opportunities
True positives	1,134	Soluble proteins correctly retained

On this earlier variant, the theoretical cost under the same weightings reaches €1,051,800, i.e. 757 × €600 + 498 × €1,200. This value measures no real laboratory cost and must not be attributed to the delivered version.

Economic threshold of the fine-tuned model: finalisation protocol

The historical Calibration partition was used to choose the epoch and early stopping of the fine-tuning. The finalisation script then split Threshold (4,052 proteins) into two distinct subsets of 2,026: Platt is fitted on the first from the checkpoint logits, then the economic threshold is chosen on the second (without retraining). The execution report confirms n = 2,026 / 2,026, with approximate separation by MMseqs2 homology groups (identity ≥ 25%, coverage ≥ 70%, capped search; not exhaustive). On the subset reserved for threshold selection, the frozen threshold is 0.296491, soluble recall 0.8575 and the simulated cost €387,600 (FN 114, FP 418; €1,200/FN, €600/FP). These are internal tuning metrics, not final independent performance figures.

The AUC ROC of 0.832857 on Tune must not be presented as independent. A retrospective export of Tune (4,052 rows) now makes it possible to recompute descriptive metrics and a confusion matrix for the fine-tuned variant at the threshold already fixed; these results are documented in Appendix F. The calibrated probabilities in

that export were reconstructed from the model's historical probabilities (inverse logit transform) then from the frozen Platt calibrator: this is not a fresh direct inference on the original logits, particularly for saturated scores. The holdout already consulted during other experiments is not an unseen cohort; no final result independent of the fine-tuning is claimed.

7. APPENDICES

Appendix D — indicators, traceability and sources

Tracking sheet to be completed during a pilot

Date / batch	Planned	Actual	Variance / action
Experiment start/end	Date, owner, objective	Observed date and duration	Delay in days
Working time	Budgeted hours	Reported hours	Workload variance
Consumables and GPU	Authorised budget	Invoices / costs recorded	Variance in € and %
Constructed/expressed	Number of experiments planned	Number and soluble fractions	Actual efficiency
Activity and purity	Criteria set	Biological results	Experimental decision

Deliverables available for review

Published application: https://huggingface.co/spaces/fatimaadda2878/protein-solubility-esm150-ft

Source code and tests: https://github.com/fatimaadda2878/protein-solubility-api

Repository documentation and evidence: README.md; docs/ARCHITECTURE.md; docs/RESULTATS_ET_LIMITES.md; docs/NOTE_EVALUATION.md; docs/PROTOCOLE_EVALUATION.md; scripts/evaluer_predictions_figees.py; resultats/evaluation_v1/ (prediction CSVs, metrics, manifest and calibration/threshold report).

Technical and academic references

- Template provided: Data & ML Project Management Report, sections 1 to 7.
- Khurana et al. (2018), DeepSol: a deep learning framework for sequence-based protein solubility prediction. Bioinformatics 34, 2605–2613.
- Lin et al. (2023), Evolutionary-scale prediction of atomic-level protein structure with a language model. Science 379, 1123–1130.
- RCSB PDB: https://www.rcsb.org/docs/programmatic-access/web-apis-overview
- UniProt API: https://www.uniprot.org/help/api
- Hugging Face Spaces: https://huggingface.co/docs/hub/spaces
- Project artefacts: Colab outputs (final campaign, UESolDS audit, threshold finalisation and 512/1,022 AA test).

The complete figures are kept here for audit; the body of the report uses rounded values suited to decision-making.

7. APPENDICES

Appendix E — Budget assumptions and verification protocol

E.1 Quantified assumptions: cumulative scenarios

Units and scopes proposed for arbitration with a laboratory; illustrative amounts excl. VAT, not committed. D = person-day. The €600 per experiment is an assumption covering consumables and platform charges for a first screen of 10 constructs (one experiment per construct), excluding scientific days already accounted for. It denotes neither a complete €2,000 cycle nor the theoretical false-positive cost of the threshold. The contingency reserve is separate from items already budgeted and does not fund the same experiments twice.

Scenario / item	Quantity	Unit price (€)	Subtotal (€)
A • audit, deduplication, data traceability	8 D	450	3,600
A • biochemistry expertise / label quality	4 D	500	2,000
A • tests and pipeline quality review	2 D	450	900
A • storage / tools / minor costs	1 lump sum	500	500
TOTAL A • data / quality	**14 D + lump sum**	**—**	**7,000**
B • reproducible evaluation, calibration	6 D	550	3,300
B • biological review and validation protocol	3 D	500	1,500
B • GPU compute / infrastructure	1 lump sum	700	700
TOTAL B • A + ML improvement	**23 D + lump sums**	**—**	**12,500**
C • running experiments and laboratory operations	5 D	500	2,500
C • screening of 10 constructs (1 experiment each, direct costs excluding labour)	10 experiments	600	6,000
C • analysis, deliverables and pilot review	2 D	500	1,000
C • contingency reserve (on justification)	1 lump sum	1,000	1,000
TOTAL C • B + biological pilot	**30 D + 10 experiments + lump sums**	**—**	**23,000**

Checks: A = 3,600 + 2,000 + 900 + 500 = €7,000; B = A + 3,300 + 1,500 + 700 = €12,500; C = B + 2,500 + 6,000 + 1,000 + 1,000 = €23,000. The €2,000 rate per complete cycle avoided is only a return-on-investment assumption and must be measured separately. Any budget decision requires quotes and validation by a biological partner.

E.2 Evaluation: traceability available and missing elements

Evidence chain published on GitHub (main branch): scripts/evaluer_predictions_figees.py (read-only metric computation, no inference); docs/NOTE_EVALUATION.md and docs/PROTOCOLE_EVALUATION.md (partition roles and limitations); resultats/evaluation_v1/predictions_tune_retrospectives.csv (4,052 input predictions); resultats/evaluation_v1/metriques_par_partition.csv and evaluation_manifest.json (metrics by length, version and SHA-256 fingerprints); resultats/evaluation_v1/bilan_calibration_seuil.json (counts 2,026 / 2,026, Platt and threshold). The script recomputes the measurements from the CSV; it performs neither a fresh ESM-2 inference nor an external homology audit. The input scores were obtained by numerical reconstruction of logits from historical probabilities: the associated calibration measurements remain exploratory and must be compared with the original logits before any claim of verified calibration.

Partition / use	Documented reality	Status of the metrics
Fit (46,270)	Training	No independent metric
Calibration (4,052)	Choice of epoch and early stopping of the fine-tuning	Consumed; do not reuse as a test
Tune (4,052)	Variant comparison: AUC 0.832857; PR-AUC 0.786022	Development, not independent
Threshold (4,052)	4,052, then two disjoint groups of 2,026: Platt (logits) and threshold selection; approximate MMseqs2 homology.	Internal tuning, not independent.
Historical holdout (4,052)	Already consulted for another variant	Not reusable as a final FT test
New cohort: to be collected	Unseen sequences, homology control, frozen protocol	Independent validation NOT carried out

E.3 Prospective validation (not carried out)

Before accessing a new cohort, pre-register the protocol and freeze the checkpoint (SHA-256), the calibrator (SHA-256), the threshold, the length policy, the experimental definition of « soluble » and the metrics. Deduplicate and audit homologies against every partition already used, documenting identity, coverage and ambiguous cases. Export identifiers, lengths, groups, labels, calibrated scores and provenance; evaluate once only, without re-optimising after observation. Report AUC ROC, PR-AUC, Brier, ECE, recall, precision and the confusion matrix with counts, for the whole set and by band ≤ 512 AA, 513–1,022 AA, > 1,022 AA. For 513–1,022 AA, recalibrate on separate data if necessary. This independent validation and the comparative biological pilot are prospects NOT yet carried out.

The PDB/UniProt search and the tag and host recommendations remain design support based on business rules: no experimental effectiveness of these recommendations is predicted or validated by the classifier.

Documentation version of 29 October 2026. The budgets are prospective assumptions; the retrospective evaluations on Tune do not constitute an independent validation.

7. APPENDICES

Appendix F — Retrospective evaluation and traceability evidence

F.1 Results documented on Tune (development)

The table below comes from resultats/evaluation_v1/metriques_par_partition.csv; it corresponds to the output of the evaluation script run read-only on a reconstructed CSV. All results concern the DeepSol Tune set already used during development. The value 0.832857 is therefore not a final measurement on unseen data. For lengths above 512 AA, these figures describe the historical truncation protocol at 512 AA, not a validation of the experimental mode at 1,022 AA.

Population / 512 AA input	N	AUC ROC	PR-AUC (AP)	Brier*
Tune — all	4,052	0.832857	0.786022	0.161064
Length ≤ 512 AA	3,672	0.834522	0.789178	0.161005
Length 513–1,022 AA	360	0.813852	0.750544	0.161934
Length > 1,022 AA	20	0.854167	0.814782	0.156193

* Brier and ECE are computed on a retrospective export of Platt probabilities reconstructed from the campaign's historical probabilities (via an inverse logit transform): they must not be presented as an externally verified calibration derived from new logits. Across the whole Tune set, ECE (10 bins) = 0.028819; recall = 0.812461; precision = 0.613128 at the historical threshold 0.296491.

F.2 Retrospective confusion matrix and simulated costs

Subset	TN	FP	FN	TP	Simulated cost (€)
Tune, n = 4,052	1,600	831	304	1,317	863,400
Threshold — threshold selection, n = 2,026	808	418	114	686	387,600

Simulated cost = €1,200 × false negatives + €600 × false positives. These two rows are not interchangeable: on Tune, the cost is a retrospective calculation after selection; on Threshold, it is the cost used to set the threshold. They are neither observed laboratory costs, nor realised savings, nor independent metrics.

F.3 Versions, files available for review and limitations

Item	Published evidence / role
Architecture	ESM-2 150M fine-tuned (last 2 layers) + Bio13; version esm2_t30_150M_UR50D_finetune_last2_bio13
Retrospective input	resultats/evaluation_v1/predictions_tune_retrospectives.csv — 4,052 rows, Tune partition
Detailed results	Published metrics file: resultats/evaluation_v1/metriques_par_partition.csv. The file predictions_auditees.csv is generated by scripts/evaluer_predictions_figees.py in the folder chosen with --sortie; it is not already committed to the repository folder.
Version manifest	resultats/evaluation_v1/evaluation_manifest.json — version, partition, threshold, metrics and SHA-256 fingerprints of the checkpoint and the calibrator.
Threshold finalisation	resultats/evaluation_v1/bilan_calibration_seuil.json — Platt on 2,026 / threshold on 2,026
Analytical reproduction	scripts/evaluer_predictions_figees.py — recomputes from the supplied CSV, without GPU or inference
Protocol and limitations	docs/NOTE_EVALUATION.md; docs/PROTOCOLE_EVALUATION.md; no external cohort evaluated

The manifest contains the SHA-256 fingerprints of the checkpoint and the calibrator, consistent with the finalisation. The script recomputes metrics from recorded predictions; it proves neither exact equivalence with the logits produced by the checkpoint, nor the independence of the proteins. External validation, calibration of the 1,022 AA mode and the economic gain remain to be established.

Source: github.com/fatimaadda2878/protein-solubility-api · 29 October 2026.